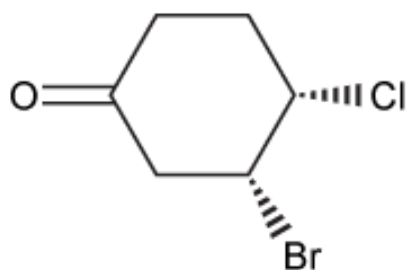
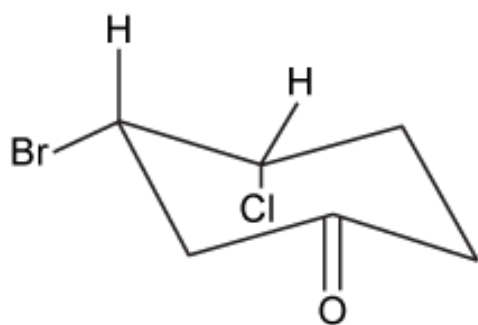


A cyclohexanone derivative with multiple substituents is shown below:



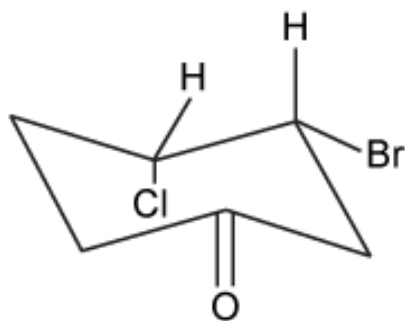
Which of the following structures depicts the same molecule?

☐ A.



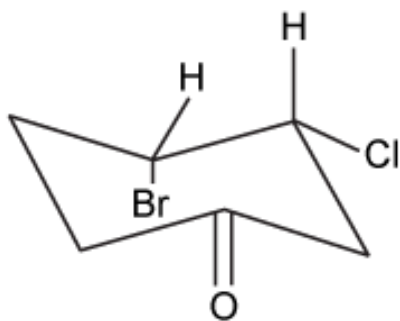
[20%]

☒ B.



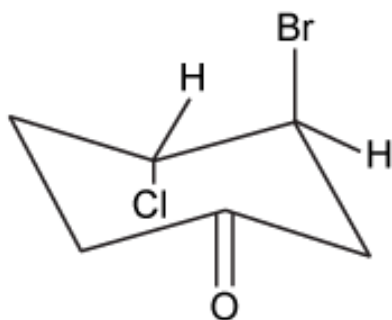
[62%]

☐ C.



[2%]

☐ D.

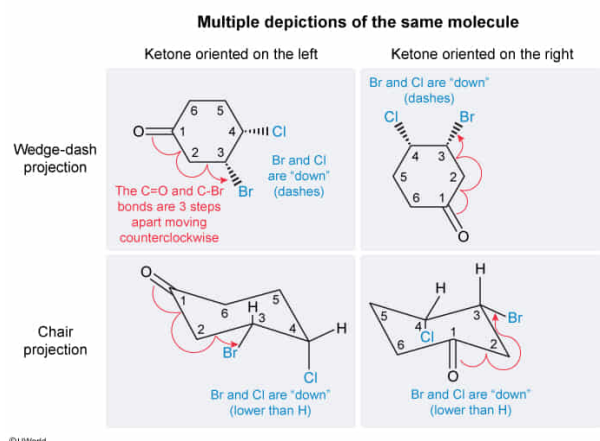


[14%]

Correct

63% answered correctly

Explanation:



A molecule can be depicted in multiple **projections** and **orientations**. For each depiction to be equivalent, all atoms within the molecule must be in the **same position relative to all other atoms**.

The question shows the wedge-dash projection of a cyclohexanone with *chloro*- and *bromo*- substituents. It is oriented with the **ketone** on the left. The C=O bond is three bonds away from the C-Br bond in the counterclockwise direction. The chlorine atom is bonded to the next carbon in the ring. Both bromine and chlorine are bonded to carbon by a dashed line, indicating that they are below or behind the plane of the ring. These features must be preserved in all depictions of the molecule.

The available choices depict the molecule as "**chair conformations**", which are commonly adopted by 6-membered rings. Instead of using dashes and wedges, chair projections show substituents in either **axial or equatorial** positions. Axial positions are perpendicular to the plane of the ring and occupy positions above or below the plane in an alternating pattern; equatorial positions alternate between slightly above and slightly below the plane. Either position (axial or equatorial) can be a dash or wedge, depending on its orientation relative to the plane of the ring. In the given molecule, both bromine and chlorine are below the plane, and the C-Br bond must be three bonds away from the C=O bond in the counterclockwise direction. Only the projection shown in Choice B satisfies all these requirements.

(Choice A) This structure depicts the C-Br bond three bonds away from the C=O bond in the *clockwise* direction, making it the **enantiomer (mirror image)** of the original molecule.

(Choice C) This structure reverses the positions of chlorine and bromine so that the C-Cl bond is three bonds away from the C=O bond in the counterclockwise direction.

(Choice D) This structure has bromine and chlorine attached to the correct carbons

but has bromine in the "up" position instead of the "down" position relative to hydrogen.

Educational objective:

A molecule can be drawn in multiple ways, with different projections and orientations. For different depictions of the same molecule to be equivalent, the atoms must be drawn in the same position relative to each other.

Time spent:0 sec

Subject:

Organic Chemistry

QID:401448

System:

Introduction to Organic Chemistry